ECON 2302 — Principles of Microeconomics

Exam 3 Review — Form A

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40 multiple-choice questions, 1 point each. Same topics as Exam 3 with new numbers, graphs, and wording — a review, not the exam. Suggested time: 70 minutes.

Name: _____ Score: _____ / 40

Question 1

Labor (workers)	Total product (units)	Marginal product	Average product
0	0		
1		4	
2			5
3	18		
4		4	
5	24		
6		2	

The above (incomplete) table provides information about the relationships between labor and various product measures. The total product that can be produced with 6 units of labor is

A. 26
B. 24
C. More information is needed to answer the question.
D. None of the above answers is correct.

Question 2

Labor (workers)	Total product (units)	Marginal product	Average product
0	0		
1		4	
2			5
3	18		
4		4	
5	24		
6		2	

The above (incomplete) table provides information about the relationships between labor and various product measures. The amount of labor that maximizes the marginal product of labor is

A. 2 units of labor.
B. 3 units of labor.
C. 4 units of labor.
D. 5 units of labor.

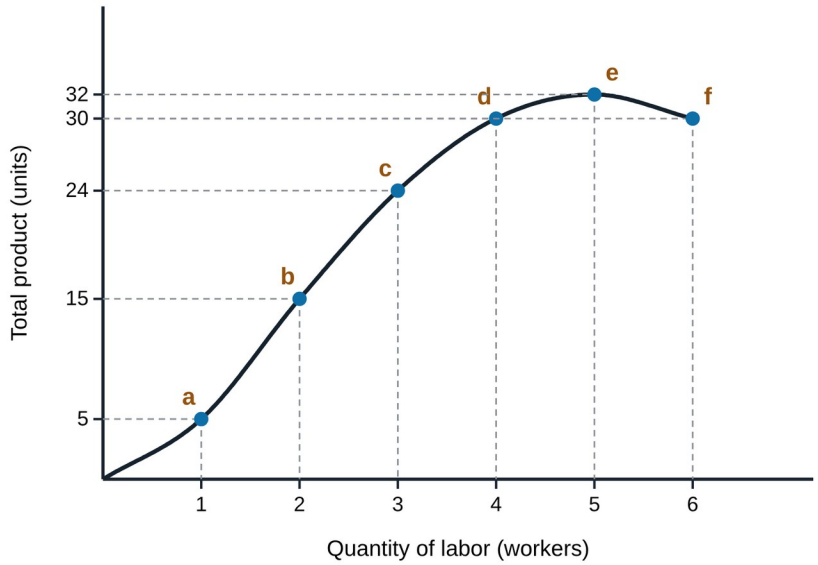
Question 3

Labor (workers)	Total product (units)	Marginal product	Average product
0	0		
1		4	
2			5
3	18		
4		4	
5	24		
6		2	

The above (incomplete) table provides information about the relationships between labor and various product measures. The average product of the fourth unit of labor

A. exceeds the marginal product of the fourth unit of labor.
B. is less than the marginal product of the fourth unit of labor.
C. is equal to the marginal product of the fourth unit of labor.
D. is equal to 4.

Question 4



In the figure above, the marginal product of the second worker is

A. 5 units
B. 7.5 units
C. 10 units
D. 15 units

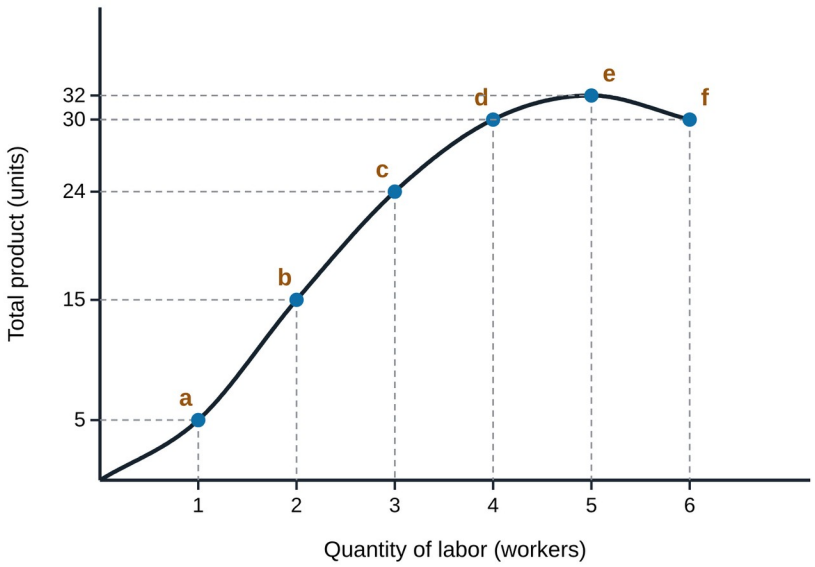
Question 5



At point d in the figure above, the average product of labor equals

A. 6
B. 7.5
C. 8
D. 30

Question 6



In the figure above, the average product of labor at point c is

A. 7.5
B. 8
C. 9
D. 24

Question 7

Labor (workers)	Output (surfboards)	Total fixed cost (dollars)	Total variable cost (dollars)	Total cost (dollars)
0	0	240		
1	24		120	
2	56			
3	66			
4	70			

The table above gives costs at Ana’s Surf Shop. Unfortunately, the record keeping has been spotty. Each worker is paid $120 a day. Labor costs are the only variable costs of production. What is the total cost of producing 56 surfboards?

A. $240
B. $360
C. $480
D. $600

Question 8

Labor (workers)	Output (surfboards)	Total fixed cost (dollars)	Total variable cost (dollars)	Total cost (dollars)
0	0	240		
1	24		120	
2	56			
3	66			
4	70			

The table above gives costs at Ana’s Surf Shop. Unfortunately, the record keeping has been spotty. Each worker is paid $120 a day. Labor costs are the only variable costs of production. What is the total fixed cost of producing 70 surfboards?

A. $240
B. $360
C. $480
D. $720

Question 9

Labor (workers)	Output (surfboards)	Total fixed cost (dollars)	Total variable cost (dollars)	Total cost (dollars)
0	0	240		
1	24		120	
2	56			
3	66			
4	70			

The table above gives costs at Ana’s Surf Shop. Unfortunately, the record keeping has been spotty. Each worker is paid $120 a day. Labor costs are the only variable costs of production. What is the total variable cost of producing 66 surfboards?

A. $240
B. $360
C. $480
D. $600

Question 10

Labor (workers)	Output (tacos)	Total cost (dollars)
0	0	1,800
5	500	3,300
10	1,300	4,800
15	1,900	6,300
20	2,400	7,800

The table above gives production and cost information for Taco Tía's. What is the total fixed cost of producing 2,400 tacos?

A. $1,800
B. $6,000
C. $7,800
D. More information is needed to determine the answer.

Question 11

Labor (workers)	Output (tacos)	Total cost (dollars)
0	0	1,800
5	500	3,300
10	1,300	4,800
15	1,900	6,300
20	2,400	7,800

The table above gives production and cost information for Taco Tía's. What is the average variable cost of producing 500 tacos?

A. $3
B. $4
C. $6.60
D. More information is needed to determine the answer.

Question 12

Labor (workers)	Output (tacos)	Total cost (dollars)
0	0	1,800
5	500	3,300
10	1,300	4,800
15	1,900	6,300
20	2,400	7,800

The table above gives production and cost information for Taco Tía's. What is the average total cost of producing 2,400 tacos?

A. $2.50
B. $3.25
C. $7,800
D. More information is needed to determine the answer.

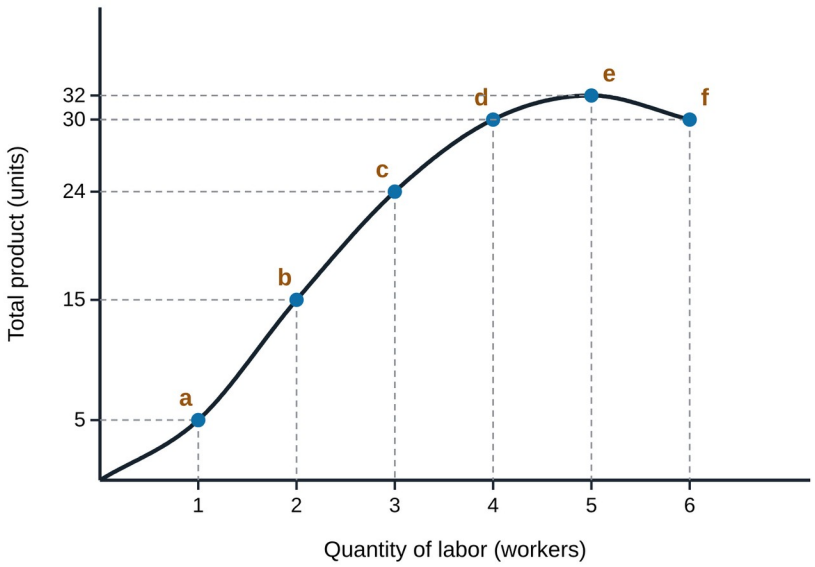
Question 13



In the figure above, after the second worker is hired, the marginal product of labor is

A. increasing.
B. diminishing.
C. constant.
D. zero.

Question 14



In the figure above, the marginal product of labor is zero at point

A. a
B. c
C. e
D. f

Question 15

Quantity sold	Price
8	$12
9	$12
10	$12

In the table above, if the firm sells 8 units of output, its total revenue is

A. $12
B. $84
C. $96
D. $108

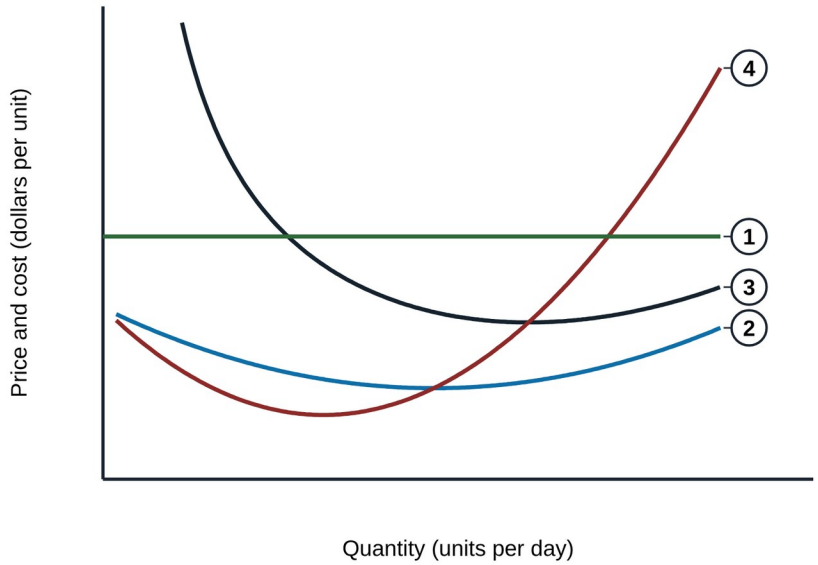
Question 16

Quantity sold	Price
8	$12
9	$12
10	$12

In the table above, if the quantity sold by the firm rises from 8 to 9, its marginal revenue is

A. $12
B. $24
C. $96
D. $108

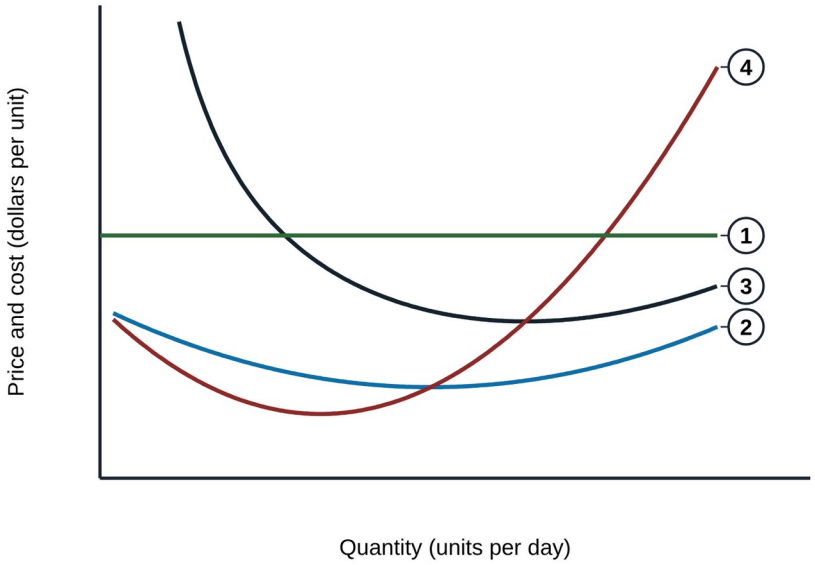
Question 17



In the figure above, the line labeled "2" is the

A. average fixed cost.
B. total cost.
C. average total cost.
D. average variable cost.

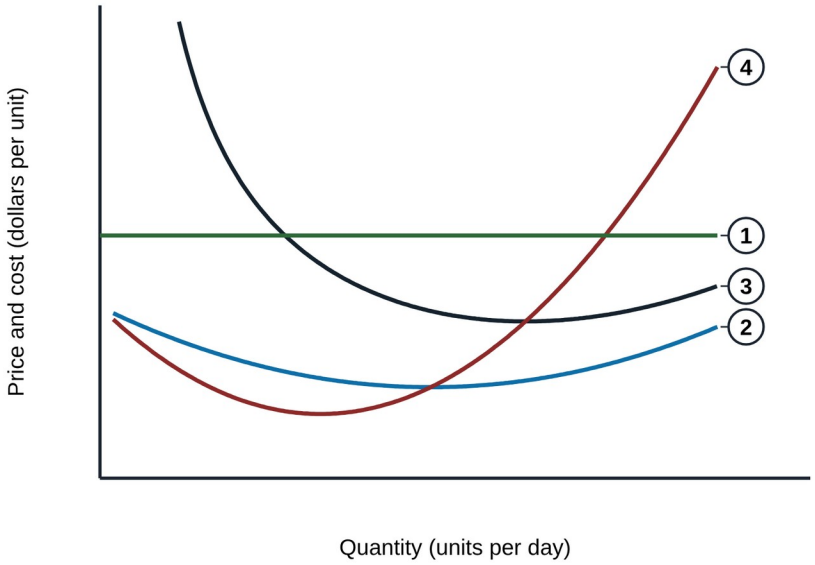
Question 18



In the figure above, the line labeled "1" is the

A. total cost.
B. average total cost.
C. average fixed cost.
D. marginal revenue.

Question 19



In the figure above, the line labeled "4" is the

A. average fixed cost.
B. average total cost.
C. marginal cost.
D. marginal revenue.

20. The short run is a time period during which

A. a firm can earn a normal profit.
B. all factors of production are fixed.
C. all factors of production are variable.
D. at least one factor of production is fixed.

21. A cost that has already been paid and cannot be recovered is called a

A. sunk cost.
B. marginal cost.
C. fixed cost.
D. variable cost.

22. A period of time long enough for a firm to vary the quantities of ALL the factors of production it uses is called the

A. variable run.
B. market period.
C. short run.
D. long run.

23. The marginal product of labor equals the

A. slope of the marginal product curve itself.
B. total product divided by the price of output.
C. total product divided by the number of workers.
D. extra output from hiring one more worker, other inputs fixed.

24. Average product of labor equals

A. total product divided by the price of the output.
B. total product divided by the quantity of labor employed.
C. the increase in output from hiring one more worker, other inputs held constant.
D. total product alone.

25. When marginal product and average product of labor are equal,

A. marginal product is at its maximum.
B. total product is at its maximum.
C. total product is zero.
D. average product is at its maximum.

26. Marginal cost is calculated as

A. total cost divided by output.
B. the change in total cost divided by the change in output.
C. the change in total cost divided by the change in labor, with capital fixed.
D. total cost minus total fixed cost.

27. As output increases, average fixed cost

A. always decreases.
B. increases, then decreases.
C. remains constant.
D. decreases, then increases.

28. Average variable cost equals

A. total cost divided by output.
B. average total cost multiplied by output.
C. the change in total cost divided by the change in output.
D. average total cost minus average fixed cost.

29. As output increases, total cost _____, total fixed cost _____, and total variable cost _____.

A. increases; increases; increases
B. increases; increases; does not change
C. increases; does not change; increases
D. does not change; increases; increases

30. Perfect competition implies that

A. there are many firms in the market.
B. all firms are price takers.
C. all firms produce an identical product.
D. All of the above answers are correct.

31. A perfectly competitive firm's demand curve is

A. a horizontal line.
B. upward sloping.
C. downward sloping.
D. a vertical line.

32. In a perfectly competitive industry, the demand for a single firm's product is

A. as elastic as the market demand.
B. inelastic, but not perfectly inelastic.
C. perfectly elastic.
D. perfectly inelastic.

33. Which of the following is ALWAYS true for a perfectly competitive firm?

A. P = AVC
B. MR = ATC
C. P = ATC
D. P = MR

34. A firm is producing the profit-maximizing amount of output when it produces where its _____ curve intersects its _____ curve.

A. MC; AVC
B. MC; TR
C. MC; MR
D. MC; ATC

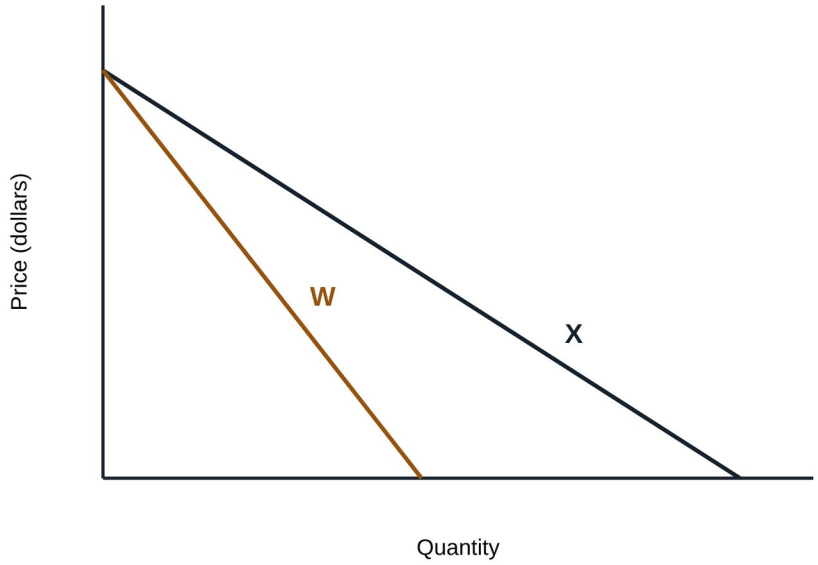
Question 35

Price (dollars per smoothie)	Quantity demanded (smoothies per day)
24	0
20	8
16	16
12	24
8	32
4	40

Sam’s Smoothies, the only seller of smoothies in a small town, faces the demand schedule shown in the table above and charges every customer the same price. Using the table, what is the marginal revenue of the 20th smoothie (the midpoint between 16 and 24 smoothies)?

A. zero
B. $4
C. $14
D. $32

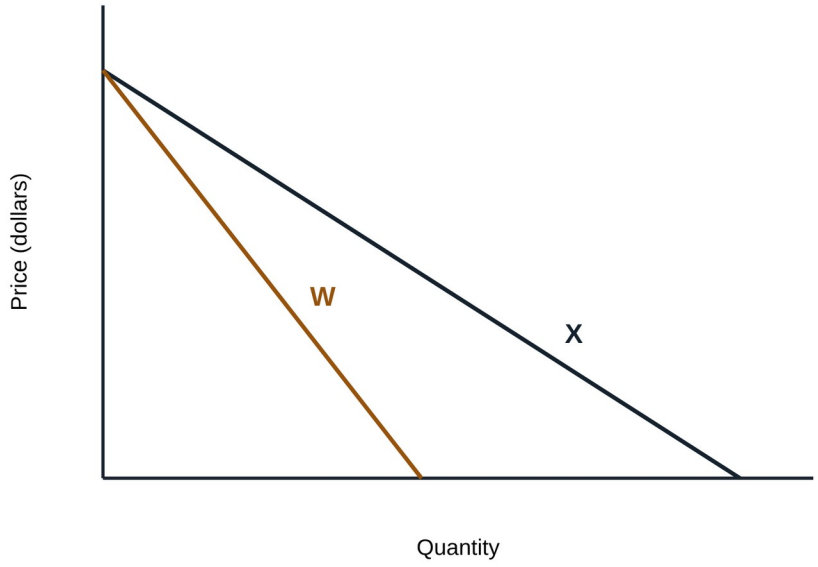
Question 36



In the figure above, X and W are the demand curve and the marginal revenue curve of the same firm. The curve labeled "X" can be

A. a monopoly’s demand curve.
B. a monopoly’s marginal revenue curve.
C. a perfectly competitive firm’s demand curve.
D. a perfectly competitive firm’s marginal revenue curve.

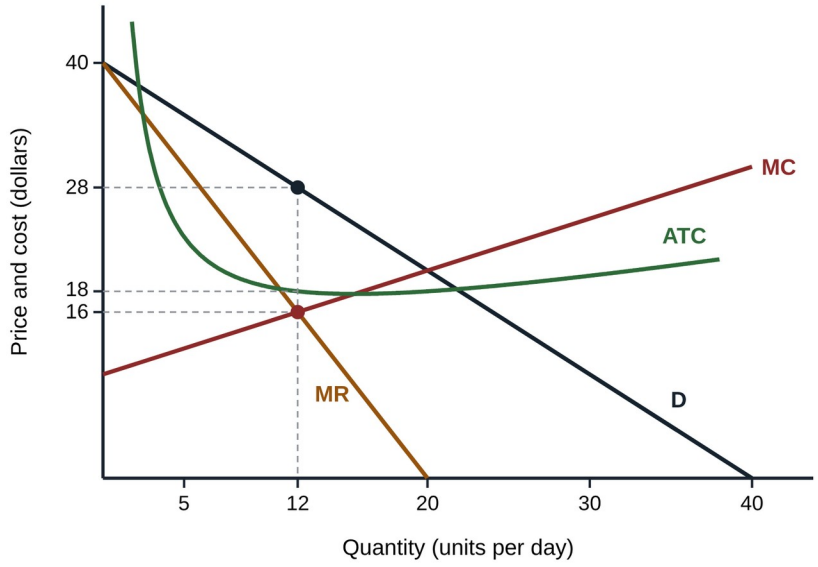
Question 37



In the figure above, X and W are the demand curve and the marginal revenue curve of the same firm. The curve labeled "W" can be

A. a monopoly's demand curve.
B. a monopoly's marginal revenue curve.
C. a perfectly competitive firm's demand curve.
D. a perfectly competitive firm's marginal revenue curve.

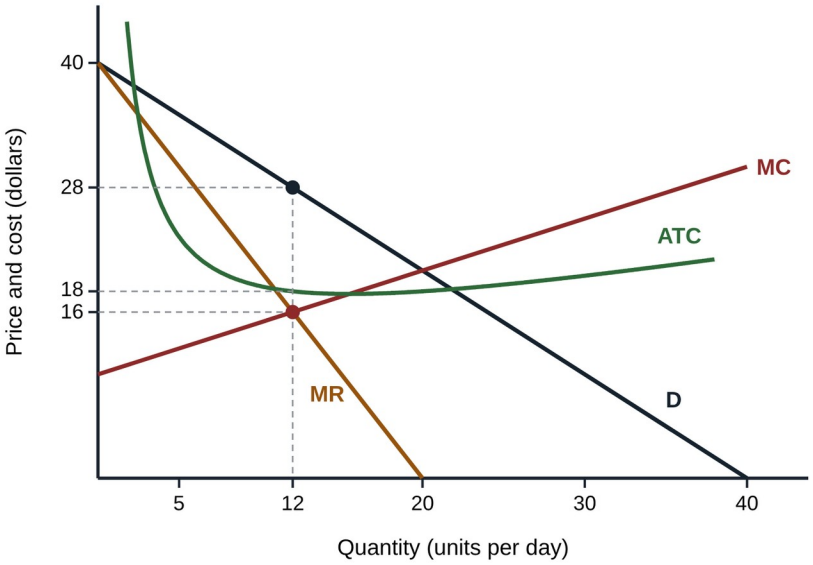
Question 38



The figure above shows the cost, demand (D), and marginal revenue (MR) curves for a single-price monopoly. The firm

A. will make an economic profit of $10 per day.
B. will charge a price of $16 per unit.
C. will produce 12 units per day.
D. will produce 20 units per day.

Question 39



In the figure above, at an output level of _____ units per day, demand is _____.

A. 5; unit elastic
B. 20; inelastic
C. 30; inelastic
D. 30; unit elastic

40. Under monopoly, when MR = 0, total revenue is

A. still increasing.
B. decreasing.
C. at its minimum.
D. at its maximum.